OpenSeeker-v2: Pushing the Limits of Search Agents with Informative and High-Difficulty Trajectories

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Research Questions

- Can simple supervised fine-tuning (SFT) alone rival the performance of complex industrial pipelines (CPT+SFT+RL) for training search agents?
- What data synthesis modifications can create high-difficulty trajectories that improve search agent performance?
- How does training data difficulty and trajectory length affect search agent capabilities?

Core Idea

The core problem is that developing state-of-the-art search agents has been dominated by industrial giants using resource-intensive pipelines (continual pre-training + SFT + RL). The key insight is that data quality trumps training complexity - specifically, using high-difficulty, information-rich trajectories can make simple SFT surprisingly effective. The approach works by scaling knowledge graph size for richer exploration contexts, expanding tool sets for broader functionality, and filtering out easy trajectories to enforce minimum difficulty, resulting in longer average trajectories (64.67 vs 36.01 steps) that force sustained reasoning.

Key Formula

Graph Expansion

$$G_{sub}^{(K)} = \mathrm{Expand}(G, v_{seed}, K)$$

G is the source knowledge graph (V,E), v_seed is a seed node, K is the expansion budget (K > k from original), and G_sub^(K) is the enlarged evidence subgraph. This formula shows how they scale graph size to create richer exploration contexts for trajectory synthesis.

Low-step Filtering

$$D_{v2} = \left\{ (q, \tau) \in D_{raw} | T(\tau) \geq T_{min} \right\}$$

D_raw is the raw trajectory dataset, τ is a trajectory, T(τ) is the number of tool-call steps in trajectory τ, T_min is the minimum step threshold, and D_v2 is the filtered dataset. This enforces a minimum difficulty floor by removing trajectories solvable in too few steps.

Key Figure

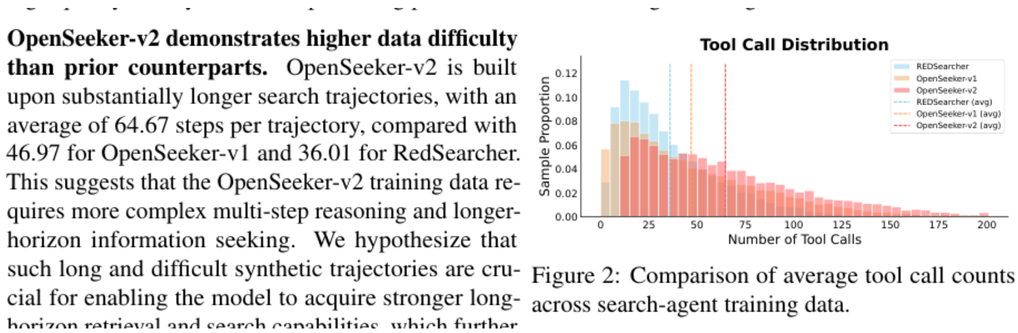
OpenSeeker-v2 demonstrates higher data difficulty than prior counterparts. OpenSeeker-v2 is built upon substantially longer search trajectories, with an average of 64.67 steps per trajectory, compared with 46.97 for OpenSeeker-v1 and 36.01 for RedSearcher. This suggests that the OpenSeeker-v2 training data requires more complex multi-step reasoning and longer-horizon information seeking. We hypothesize that such long and difficult synthetic trajectories are crucial for enabling the model to acquire stronger long-horizon retrieval and search capabilities, which further

Figure 2: Comparison of average tool call counts across search-agent training data.

Figure 2: Shows tool call distribution comparison demonstrating OpenSeeker-v2 uses substantially longer trajectories (64.67 avg steps) than prior methods, validating that higher data difficulty drives performance gains.

Limitations

- Only evaluated on 30B parameter models within ReAct paradigm - scalability to other architectures unclear
- Training data is synthetically generated from knowledge graphs which may not cover all real-world search scenarios
- Limited to 10.6k training samples - unclear if approach scales with more data
- No analysis of computational cost during inference for longer trajectories